

Astronomers want telescopes in space because the Earth's atmosphere creates significant obstacles for ground-based observation.

Space telescopes are incredibly difficult and expensive to build and deploy but this allows scientists to achieve remarkable scientific results for two main reasons.

1. They remove atmospheric blurring. Earth's atmosphere is always churning and a constantly changing temperature, wind condition combination causes seeing. It is the blurring and "twinkling" of stars and the result of the moving air distorting incoming starlight which makes even a perfectly designed ground-based telescope have a limit of sharpness dictated by atmospheric seeing. When a space telescope such as the Hubble Space Telescope is placed above the atmosphere this limit of seeing is removed and the telescope mirror is said to have reached its diffraction limit. This has the effect of the images produced by ground-based telescope being "mushy" compared to those produced from space.

2. They overcome atmospheric opacity. The Earth's atmosphere provides a blanket around the Earth and this is all very well for life on Earth but not so great for astronomy. Most wavelengths of the electromagnetic spectrum such as X-rays, gamma rays and most UV radiation are absorbed and cannot make it to ground level. In order to see the Universe at these wavelengths our detectors must go into the vacuum of space.

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Though space telescopes can be incredibly difficult and costly to build and operate, they are indispensable tools for many areas of astronomy and science in general. This is because there are two primary advantages to placing a telescope above Earth's atmosphere:

1. *Overcoming Atmospheric blurring:* Light from stars twinkle in space, due to heat and air movements around the earth, or "seeing". This is one of the limiting factors for ground-based telescopes, even if the telescope itself is optically perfect. In orbit, above Earth's atmosphere, space telescopes, like the Hubble Space Telescope, can achieve the "diffraction limit" of the mirror, yielding much clearer and sharper images than ground-based telescopes.

2. *Overcoming atmospheric opacity:* The atmosphere around the Earth provides many advantages to our way of life, but is a nuisance for astronomers. Nearly every type of electromagnetic radiation that is not visible light or some of the radio waves is absorbed by Earth's atmosphere and will not reach us here on the surface. Therefore, to observe the universe in X-rays, gamma-rays, and ultraviolet light, we must do so in the void of space.

Examples of wavelength regions beyond the spectrum of visible light where astronomers can learn about the universe include the radio and X-ray regions.

By extending our view beyond what our eyes are capable of seeing, we can examine physical processes that are completely obscured in the visual spectrum.

Radio waves are light with the longest wavelength. Because radio waves are capable of penetrating immense clouds of interstellar dust which obscures objects in the visible spectrum, the use of radio telescopes gives us a tool with which we can probe the center of galaxies and cold gas where stars are born. Furthermore, these high-energy phenomena include objects such as pulsars and the leftover shells of destroyed stars.

X-rays are light with the shortest wavelength and highest energy. Through X-ray astronomy we can observe the most violent and hottest phenomena in the universe. As X-rays are released by objects which are billions of degrees, this wavelength is capable of detecting such phenomena as gas streaming into a black hole or colliding clusters of galaxies and extreme environments surrounding neutron stars.

Through the combination of large apertures, vantage points above the Earth, and multi-wavelength astronomy, astronomers have moved from a universe viewed as dots of light to one viewed as a complex, high-energy object.